open research problems

Diagnosis-driven task invention

Learning coupling structure

Separate information gain from population shift

Evaluator co-evolution

Long-horizon reproducibility

Learning to learn, measured

Questions for the *ACL community

Benchmarking self-improvement. Streams with fixed targets and evolving tasks, where adding easy tasks cannot inflate progress (sketch, right).

Multilingual and domain adaptation. Methods built on English and high-resource benchmarks may not transfer, and feedback loops can widen gaps between languages.

Online versus offline. Which updates belong in deployment, and which in a curated offline cycle? All three case studies learn offline or between rounds.

Reward hacking and proxies. How to keep an agent from optimizing its judge instead of its task. Cursor’s real-time RL already met this in production.

Discussion: which of these would you put in a shared task?

A design and evaluation checklist for self-improving agents

Design commitment	Evidence or specification required
Target tasks	Inputs, outputs, populations, metrics, protected constraints
Learner boundary	Persistent state, transient context, external dependencies
Experience	Sources, selection process, supervision, exposure history
Retained pathway	What changes and how it can affect future decisions
Views	Complementarity, provenance, dependence assumptions
Coupled tasks	Meaning and permissions of each learning edge
New tasks	Diagnosis, specification, alternatives, target transfer test
Substrates	Scope, generalization, cost, auditability, reversibility
Governance	Proposal, assessment, authorization, commitment, monitoring
Repair	Version bundles, descendants, rollback, compensation
Evaluation	Frozen and updating comparisons, prospective tests, uncertainty
Reporting	Gains, failures, resources, limitations, reproducible history

The conclusion: conditional and testable

- A deployed language agent can improve through retained changes across many substrates: knowledge bases, wikis, skills, harness code and weights.

- That improvement is established only when future performance, learning efficiency or preservation under change improves under a defensible comparison.

- What makes it credible is evidence, governance, revision and longitudinal evaluation.

Where to read next

- Start with the foundations: Mitchell et al. define never-ending learning; Silver and Sutton argue for agents that learn from their own stream of experience.

- Add the two literatures this tutorial connects. Continual learning of LLMs, which Hou et al. describe as moving from weights to memory, skills and protocols; and self-evolving agents with persistent memory.

- Check also related tutorials. Two 2025 tutorials treat continual learning of LLMs mainly at training time, one at IJCNLP-AACL 2025. This tutorial adds the post-deployment, governed, multi-substrate view.